CIS 678 - Machine Learning

Predictive modeling: Classification

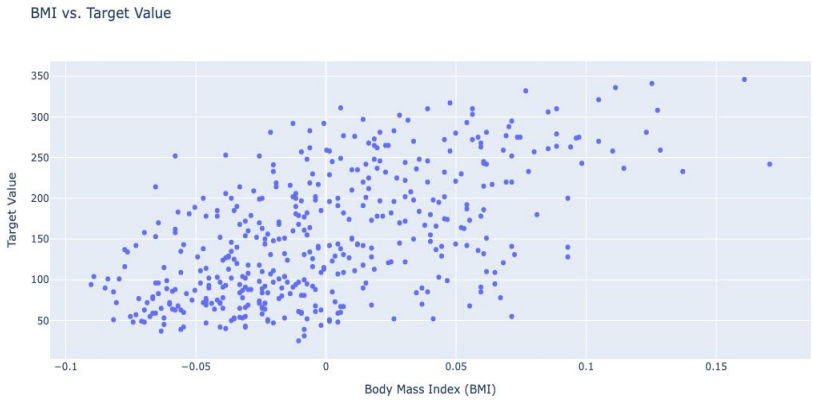
Regression vs Classification

- Essentially, we are learning some relationship among variables.

$$f(y|X)$$

Regression vs Classification

- Essentially, we are learning some relationship among variables.



$$f(y=diabetes\text{-}score|X='bmi')$$

Regression vs Classification

- Essentially, we are learning some relationship among variables.

$$f(y|X)$$

*The difference is with the variable, **y***

Regression problems

$$y \in R$$

- *Insurance cost :* $y \in R$
- *House price:* $y \in R$
- *Weather prediction:* $\boldsymbol{Y \in R}$
- *Energy consumption:* $y \in R$
- *Sales forecasting:* $y \in R$

We have learned about regression
(not complete; will continue ..)

Regression vs Classification

$y \in R$

- *Insurance cost : $y \in R$*
- *House price: $y \in R$*
- *Weather prediction: $\mathbf{Y \in R}$*
- *Energy consumption: $y \in R$*
- *Sales forecasting: $y \in R$*

We have learned about regression
(not complete; will continue ..)

$y \in \{....\}$

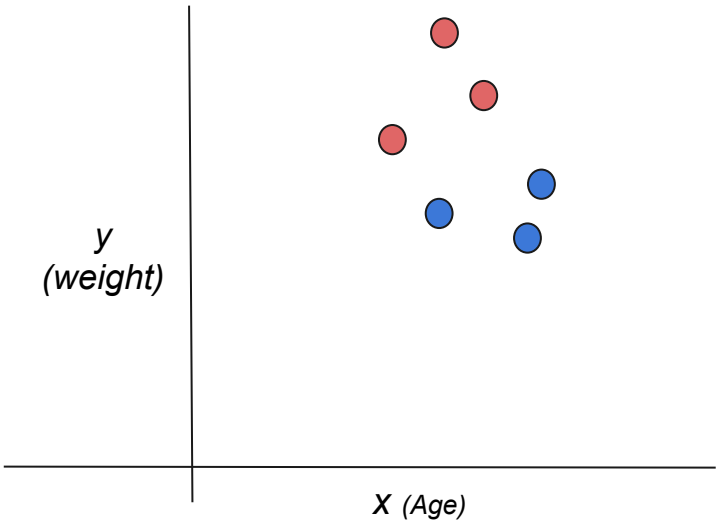
- *Cat vs dog: $y \in$ {cat, dog}*
- *Spam filter: $y \in$ {spam, not-spam}*
- *Digit classification: $y \in \{0,1,....9\}$*
- *Sentiment classification: $y \in$ {happy, sad, confused, angry …}*

note: has to be from a closed set

We are starting today

Classification Modeling

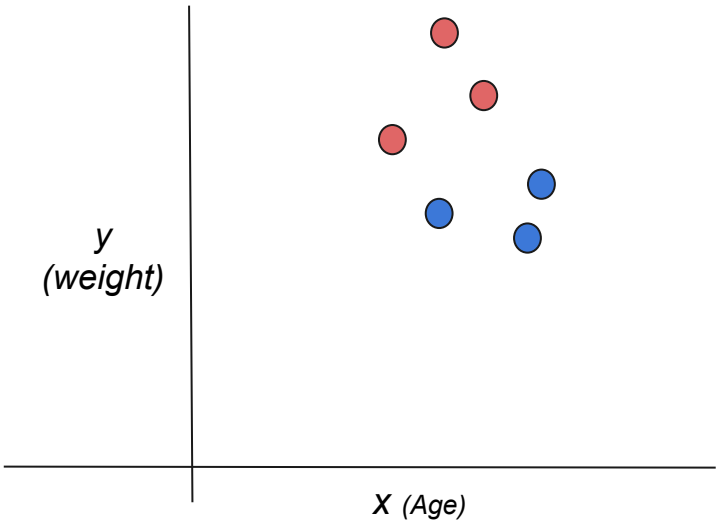
Linear Line as a Threshold



- We have some overweight and underweight people

$$y \in \{underweight(0), overweight\ (1)\}$$

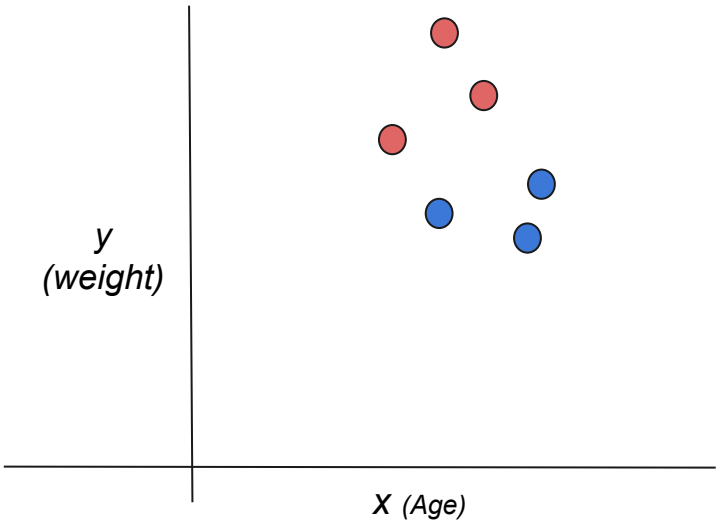
Linear Line as a Threshold



- We have some overweight and underweight people
- We need to learn a classifier

$$y \in \{underweight(0), overweight\ (1)\}$$

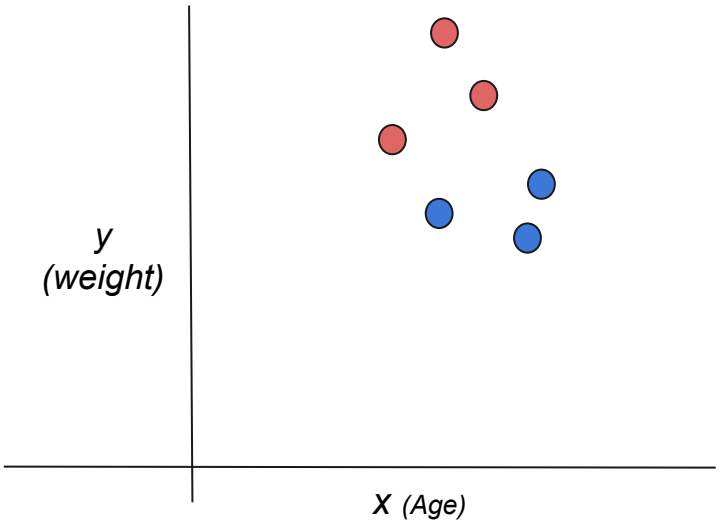
Linear Line as a Threshold



- We have some overweight and underweight people
- We need to learn a classifier
- Can we do it using a straight line?

$y \in \{underweight(0), overweight\ (1)\}$

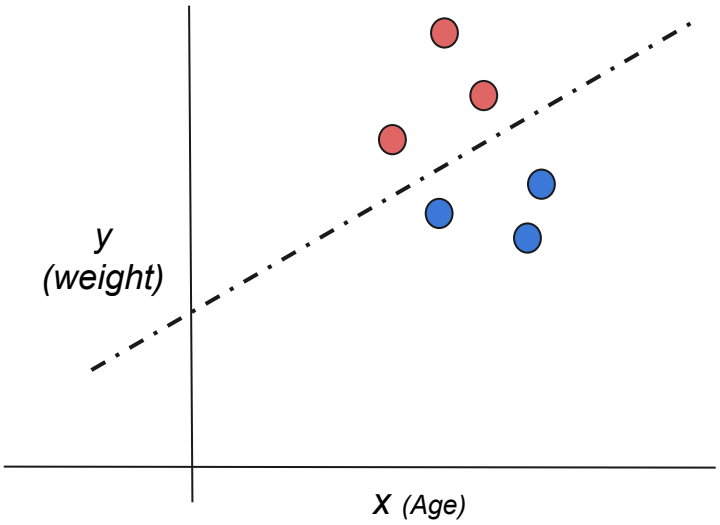
Linear Line as a Threshold



- We have some overweight and underweight people
- We need to learn a classifier
- Can we do it using a straight line?
- Yes; how ..

$$y \in \{underweight(0), overweight\ (1)\}$$

Linear Line as a Threshold

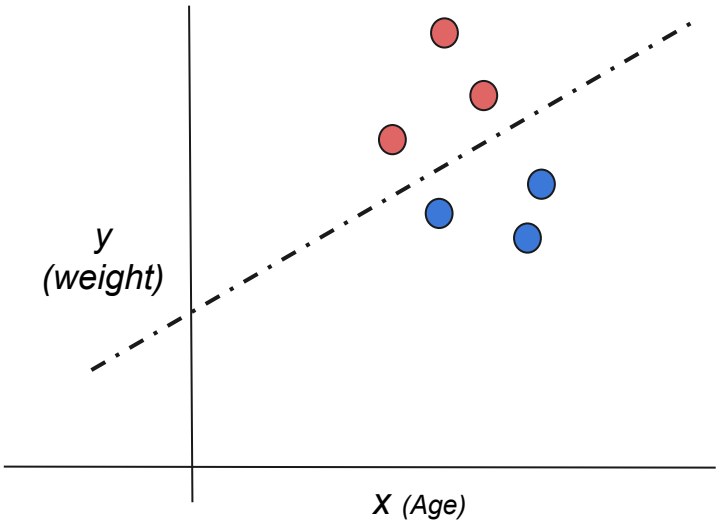


Straight Line as the Separator

$$\hat{y} = \beta_0 + \beta_1 x$$

y ∈ {underweight(0), overweight (1)}

Linear Line as a Threshold



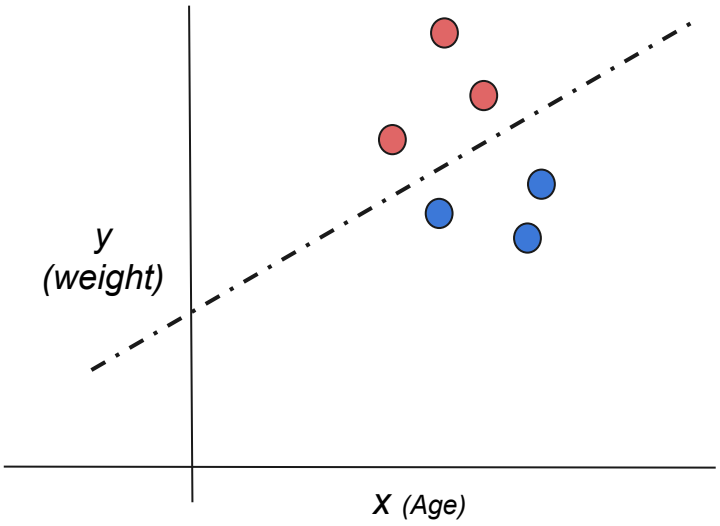
Straight Line as the Separator

$$\boxed{\hat{y} = \beta_0 + \beta_1 x}$$

Classification Rule

$$\boxed{\check{y} = \begin{cases} 1, & \text{if} \quad y - \hat{y} > 0 \\ 0, & \text{otherwise} \end{cases}}$$

Linear Line as a Threshold



Straight Line as the Separator

$$\boxed{\hat{y} = \beta_0 + \beta_1 x}$$

Classification Rule

$$\check{y} = \begin{cases} 1, & \text{if} \quad y - (\beta_0 + \beta_1 x) > 0 \\ 0, & \text{otherwise} \end{cases}$$

Classification Models

- Logistic Regression
- k-NN
- Decision Tree
- Random Forest Classifier
- Support Vector Machines (SVMs)
- Boosting Classifiers
- Naive Bayes

QA